

Steady-State Solution

Fixed preferences and tax rules; no immigration.

1. Guess the house price P , pension ϖ , and rebate T ; derive rent r from asset pricing.
2. Solve households backward over age: optimize saving and housing conditional on fertility and tenure, then compute choice probabilities.
3. Construct the stationary household composition. Set household mass equal to housing supply divided by mean housing per household.
4. Update P , ϖ , and T until household entry replaces exits, payroll taxes fund pensions, and property taxes fund equal rebates.
5. Apply the population law for one period and verify that the distribution and entry queue reproduce themselves; check markets and both budgets.

Transition Solution

Permanent preference change with a stationary terminal condition.

1. Fix inherited households and the entry queue; solve the new steady state for the terminal value.
2. Guess price, pension, and rebate paths (P_t, ϖ_t, T_t) ; derive rents from asset pricing.
3. Using these paths, solve households backward in calendar time.
4. Advance households and entry queues; aggregate housing demand and taxes.
5. Update all three paths jointly until housing clears and both fiscal budgets balance at every date.
6. Check terminal distributions and queues against the steady state; extend the horizon to test stability of early responses.